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Decision-making guide

Value Chain

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes market, Demand, Production, Marketing, finance and finally the Decision checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

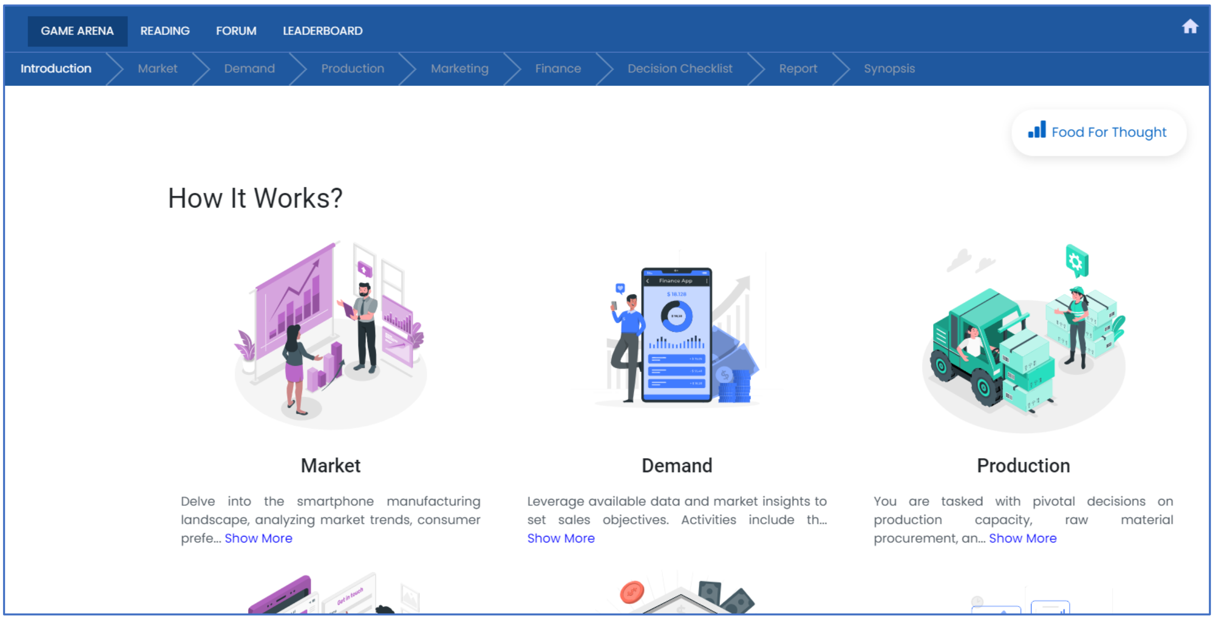

Figure 1 The Game Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the industry, ongoing trends, obstacles, and the diverse array of decisions required.

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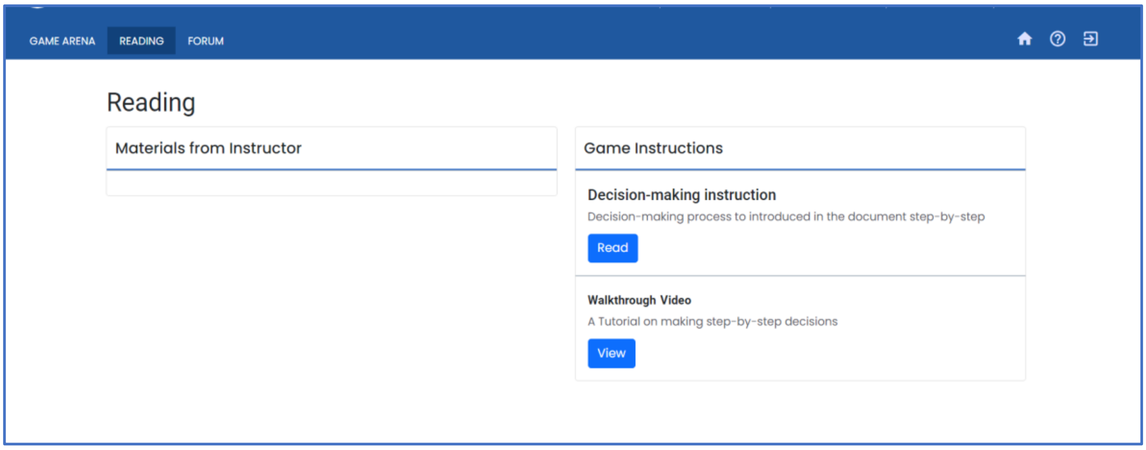

Figure 2 The Readings Section

1.3 Forum

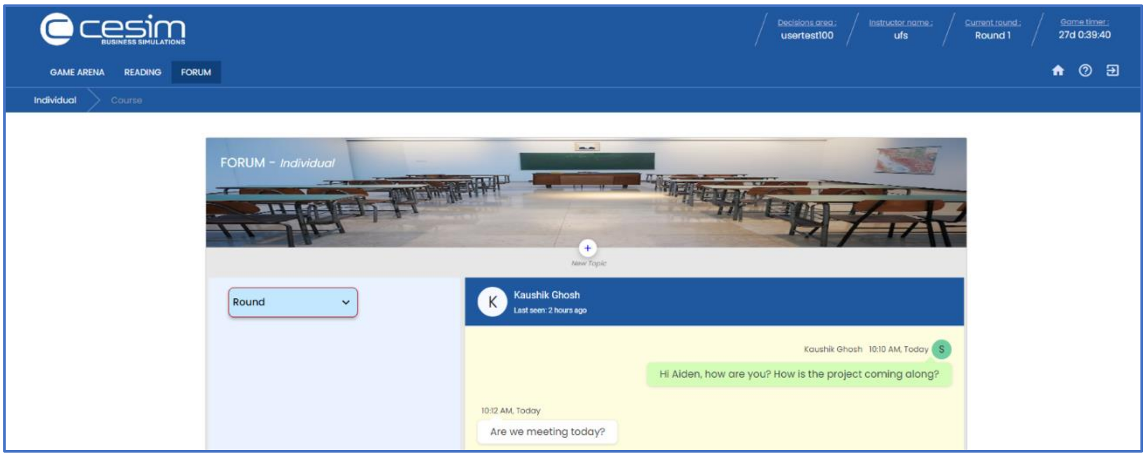

Figure 3 The Forum Section

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

In the value chain simulation, participants assume the role of a strategist at Mobile Marvels, a company strategically positioned in India's dynamic mid-range smartphone market. With a focus on tech-savvy consumers seeking affordable yet feature-rich devices, Mobile Marvels competes robustly against twelve major players. The company distinguishes itself through its dedication to versatile features and localized customization tailored to diverse regional preferences.

Participants in the simulation oversee critical areas such as demand forecasting, production planning, investment strategies, marketing initiatives, and financial management. These decisions collectively span the entire business value chain, aiming to achieve a market share of 8.3% or higher and maintain an operating margin exceeding 10%.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

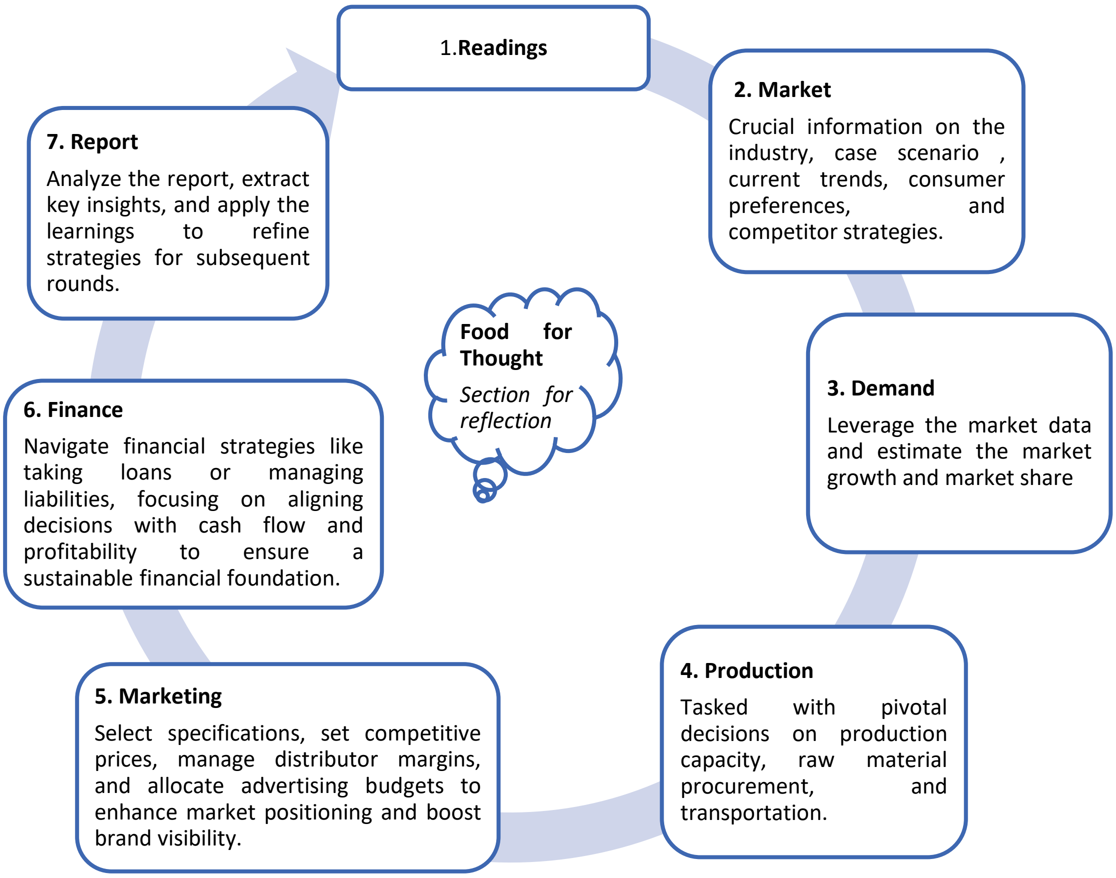

Figure 4 The Sequence of Decision-Making Process

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2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab’s decision-making.

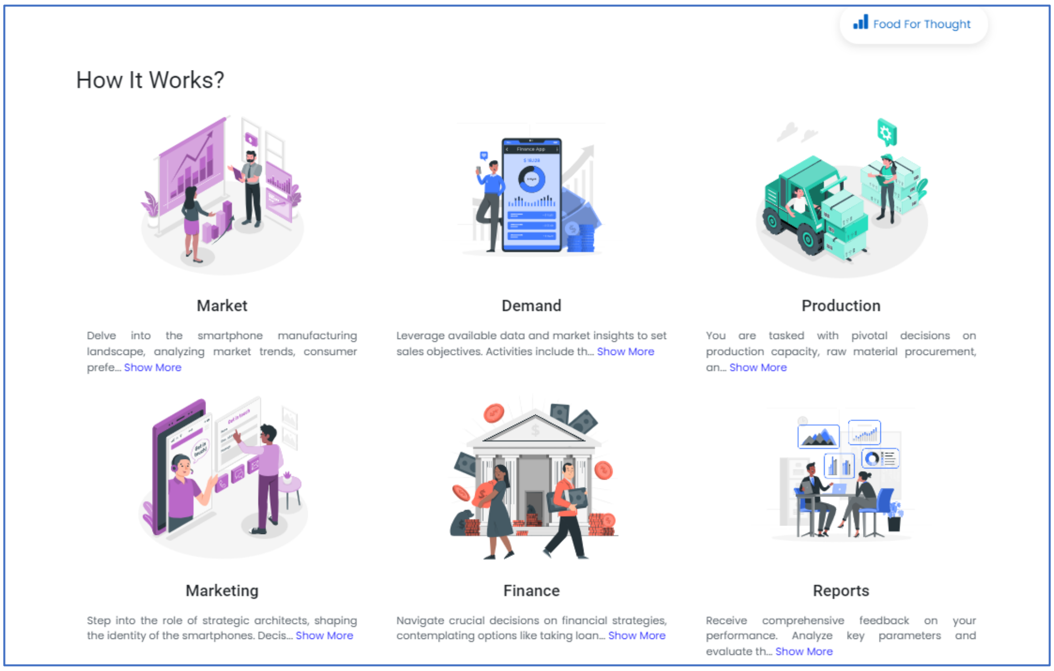

Figure 5 The Introduction Tab

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making

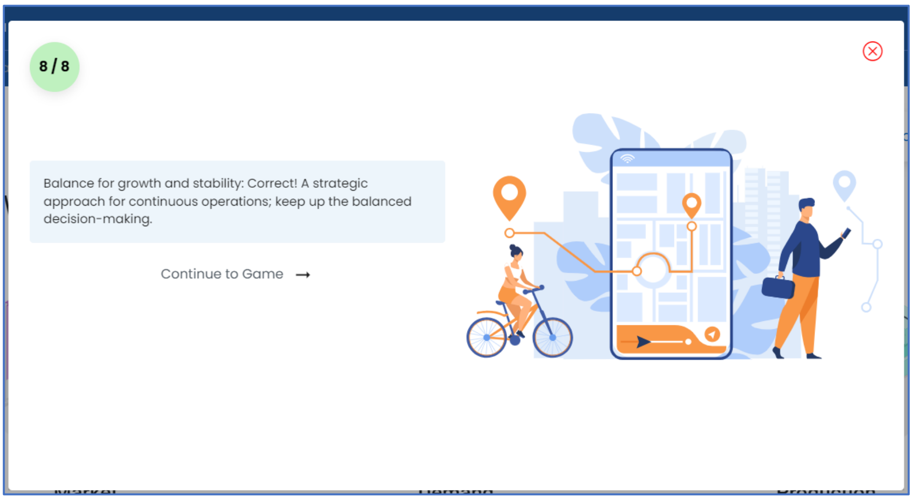

Figure 6 The Food for Thought Section

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2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on case scenario, industry overview and the key parameters to consider when making decisions in each tab.

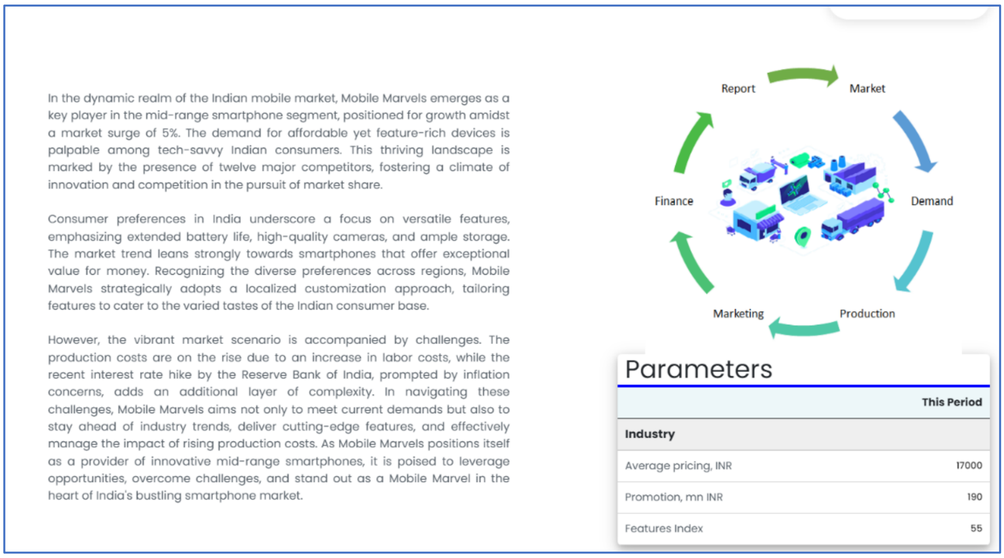

Figure 7 The Market Tab

2.4 Demand

The decision-making process starts with the demand section, where participants need to forecast expected market growth of the industry concerned and their anticipated market share for the current round. Historical data, including previous period market volume, market size, market share, and competitor numbers, as well as insights from the market section supports participants in making these estimates. Participants estimate the expected expansion of the market and predict the portion of the market they aim to capture.

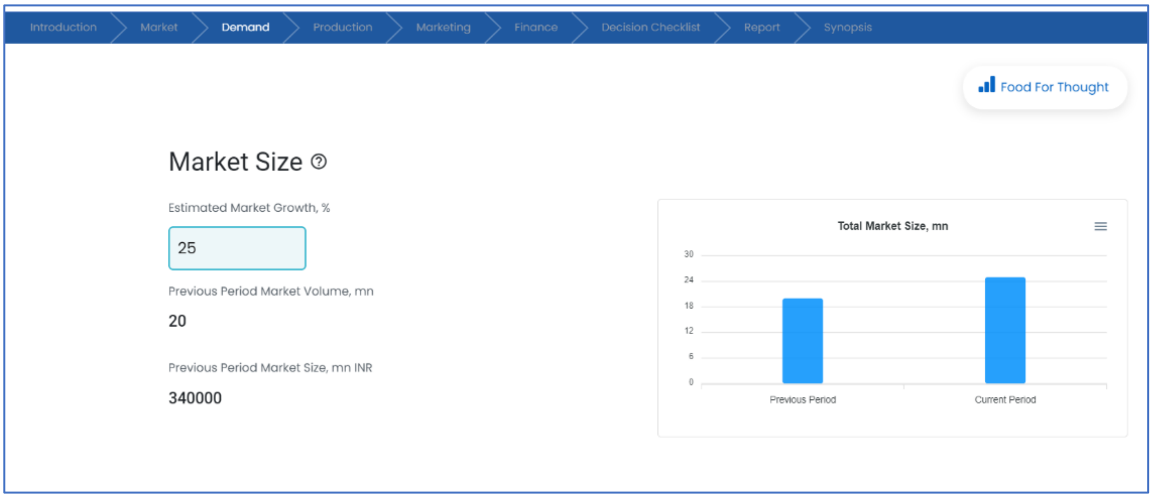

Figure 8 Market Size decision area under the Demand Tab in game Arena

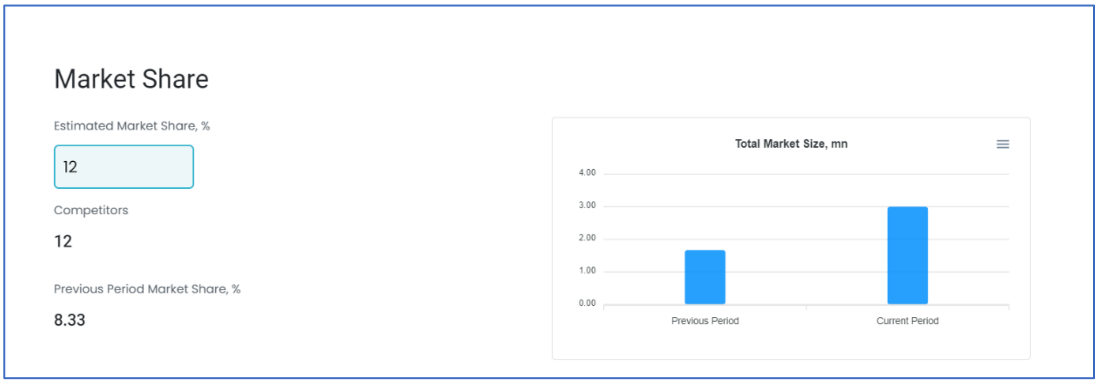

Figure 9 Market Share Decision area under the Demand Tab in Game Arena

2.5 Production

Decision 1 - Production Planning

Production planning is a critical aspect of managing the efficient production of goods to meet market demand while optimizing costs and resources. Participants in this process are tasked with planning the production of estimated demand through two primary methods:

1. **In-house Manufacturing**: This method involves using the company's own facilities and workforce to produce goods.

2. **Outsourcing:** This method involves contracting external suppliers to produce goods. It can provide flexibility, reduce capital expenditure, and leverage specialized expertise, but it may also involve risks related to quality control, supply chain disruptions, and less direct oversight.

The decision-making process involves allocating production capacity between these two methods. The production cost per unit will be adjusted based on the capacity allocated to each method. Participants must balance these methods by considering production costs, meeting estimated demand, and the maximum capacity for each production method.

The production planning graph is a valuable tool for understanding inventory levels, including initial inventory, production capacity, outsourcing capacity, demand fulfilment, and the potential for closing inventory or stock-outs. This graph aids in strategic decision-making to ensure efficient and cost-effective production planning.

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Figure 10 The planning area under the Production Tab in Game Arena

Decision 2 – Investment

To enhance production capacity, participants can invest in additional machines that boost production capabilities. The costs of acquiring these new machines must be factored into the overall increase in total production capacity.

Figure 11 Plant Investment Decisions under Production Tab in the Game Arena

Decision 3 – Component Supplier

Participants also have the option to select the optimal supplier for delivering components. Each vendor's advantages, disadvantages, and associated costs are outlined in the decision area to facilitate informed decision-making.

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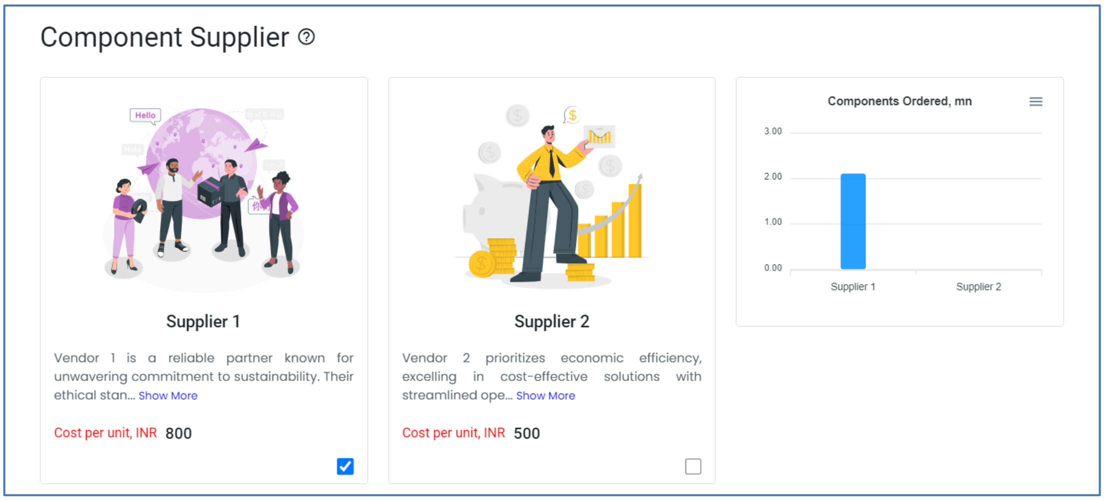

Figure 12 Decision on Supplier under the Investment tab in Game Arena

Decision 4 – Transportation

In this section, participants are tasked with selecting an appropriate transporter based on factors such as cost, reliability, and experience.

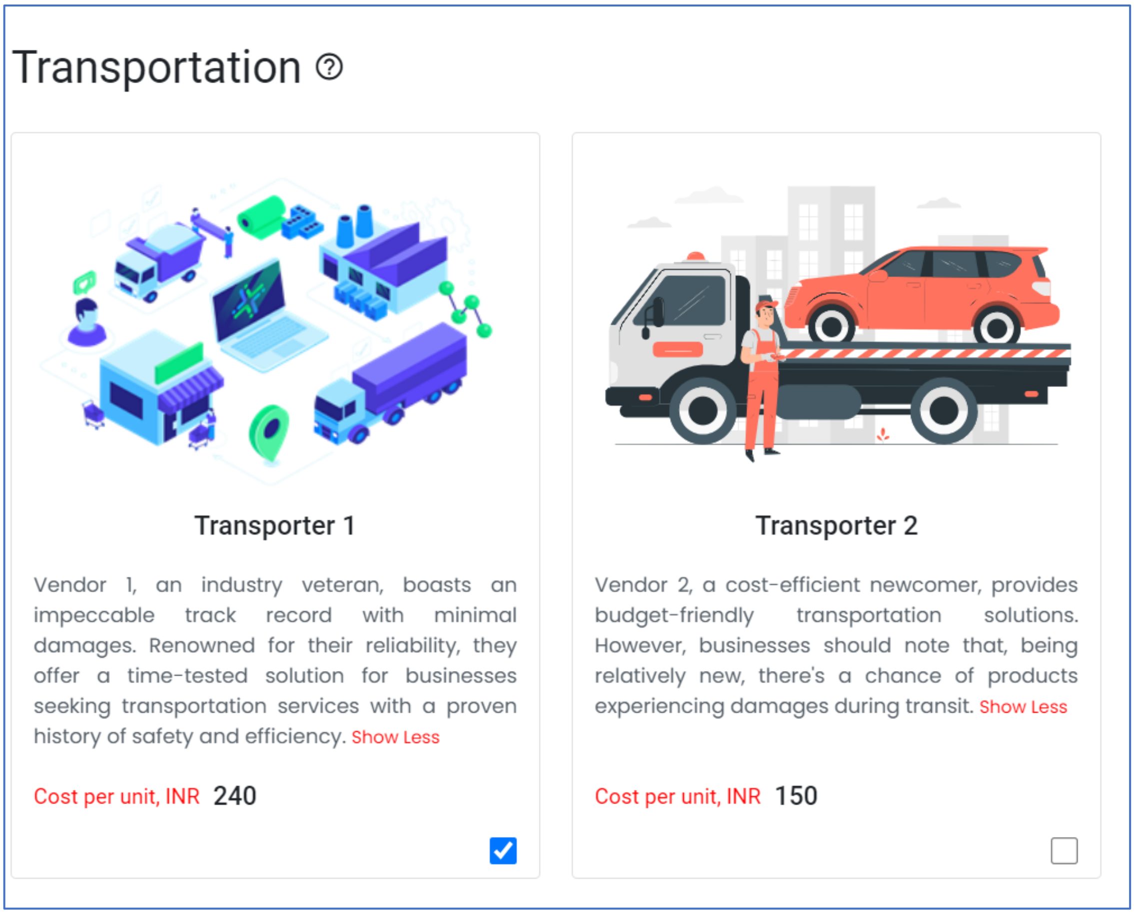

Figure 13 Decision on Transportation under the Investment tab in Game Arena

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2.6 Marketing

Decision 1 - Marketing Mix

Under the marketing decision tab, participants are tasked with deciding the marketing mix that includes the feature index of the product, promotion budget, as well as price.

a. **Feature Index:** The set of product attributes and specifications that appeal to consumers, including quality, design, functionality, and additional features. A higher feature index attracts consumers seeking high-quality, innovative, or premium features. Enhancing these features helps the product stand out in a crowded market and allows for targeted marketing strategies by appealing to different market segments that value specific features. While deciding on the feature index, consider market elasticity and cost implications.
b. **Promotion:** Under the promotion decision area, participants are tasked with allocating a budget towards marketing activities. This aims to increase product awareness and sales, boost brand visibility and consumer engagement, making it essential for market penetration.

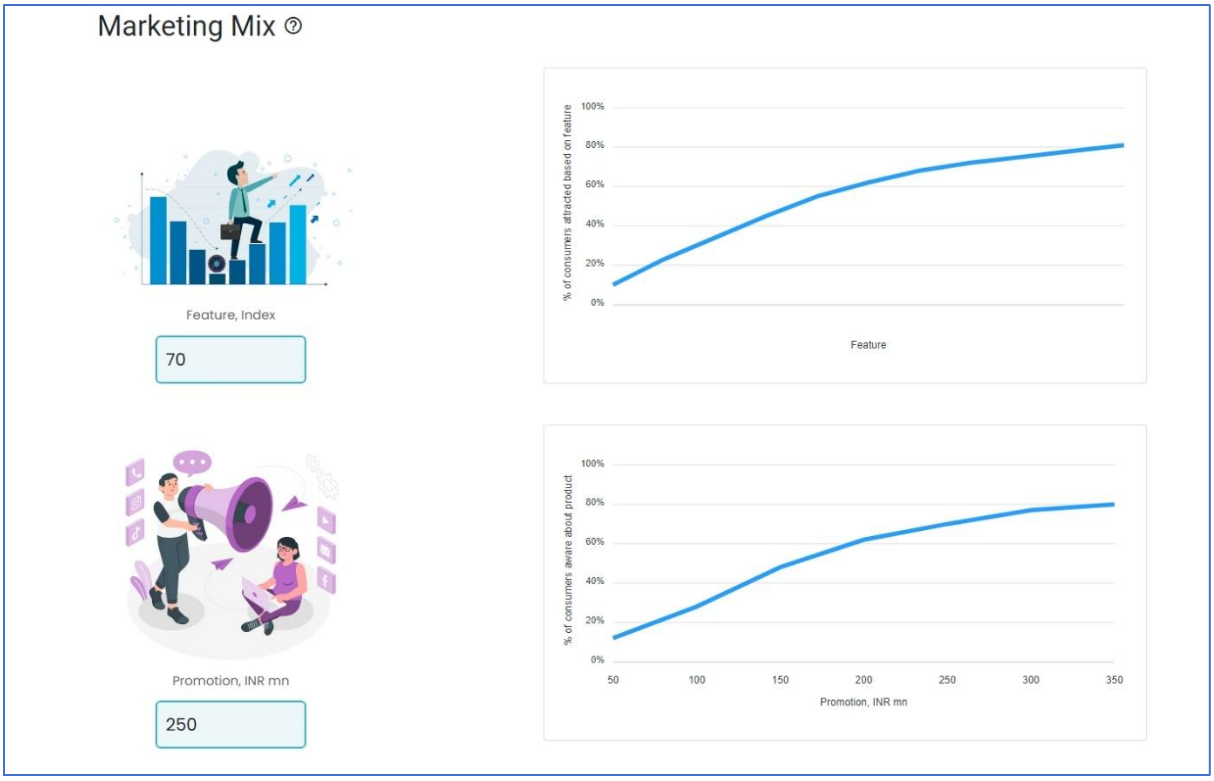

Figure 14 Marketing Mix Decisions under the Marketing Tab in Game Arena

c. **Price**: Defined as the amount consumers are willing to pay for a product, playing a crucial role in the marketing mix due to its direct impact on sales volume and profitability. Price influences consumers' perception of a product's value and quality, where a higher price often implies premium quality and exclusivity, while a lower price suggests affordability. Understanding price elasticity helps predict how changes in price will affect consumer demand, crucial for pricing strategies. When making pricing decisions, it's essential to analyse how consumers respond to different price points, ensuring competitive pricing aligned with market conditions, while also maintaining adequate profit margins to sustain business growth and profitability.

Figure 15 Marketing Mix Decisions under the Marketing Tab - Game Arena

Decision 2 - Distributor Margin

Participants are tasked with determining the optimal margins for online distributors, specialty stores, and large retailers to maximize reach while aligning with the firm's pricing strategy. Each distribution channel serves a distinct strategic purpose. Margin decisions must carefully consider the role of each channel in reaching target markets, enhancing brand perception, and supporting overall profitability within the broader pricing framework.

1. **Online Channels:** The distributor margin for online channels ranges from 5% to 10%. These channels offer broad distribution and cost efficiencies, leveraging lower operational costs to support competitive pricing and potentially drive higher sales volumes.

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2. **Specialty Stores:** Specialty stores, focused on niche markets, are offered margins ranging from 7% to 15%. These stores enhance brand perception and can command premium prices by providing personalized customer service and a carefully curated product selection.

3. **Large Retailers:** Large retailers provide wide-scale distribution with distributor margins ranging from 5% to 12%. By leveraging their extensive networks, they enhance brand visibility and attract a diverse customer base, focusing on convenience and accessibility.

Decision 3 – Innovation & Services

Here in this section, participants are tasked with choosing innovation and services. By strategically incorporating innovations like predictive analytics, subscription-based services, blockchain integration etc., company can achieve a multi-faceted impact. These initiatives not only enhance consumer perception and operational efficiency but also position the company for sustained growth and competitiveness in the industry. Participants should carefully evaluate each innovation's potential to align with their business goals and customer expectations, ensuring a balanced approach that maximizes both short-term gains and long-term strategic benefits.

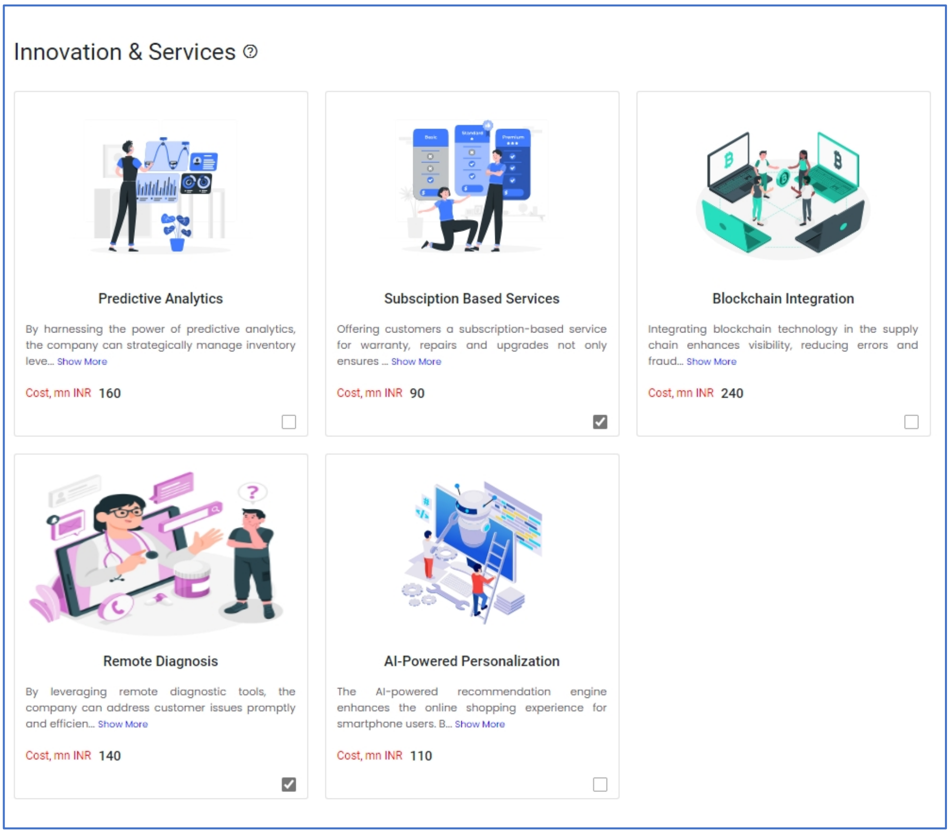

Figure 16 Innovation & Services under the Marketing tab in Game Arena

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2.7 Finance

In the finance decision area, participants can access comprehensive data on the company's financial health, including initial cash, cash generated from operations, investments in plants, cash obtained from borrowings, and ending cash balance. In cases where there is a cash shortfall, participants have the option to secure long-term loans to bridge the gap.

If the company runs out of cash and cannot meet its cash requirements, the system will automatically initiate emergency loans. It is important to note that these emergency loans typically carry higher interest rates than standard long-term loans, potentially affecting the company's overall financial sustainability.

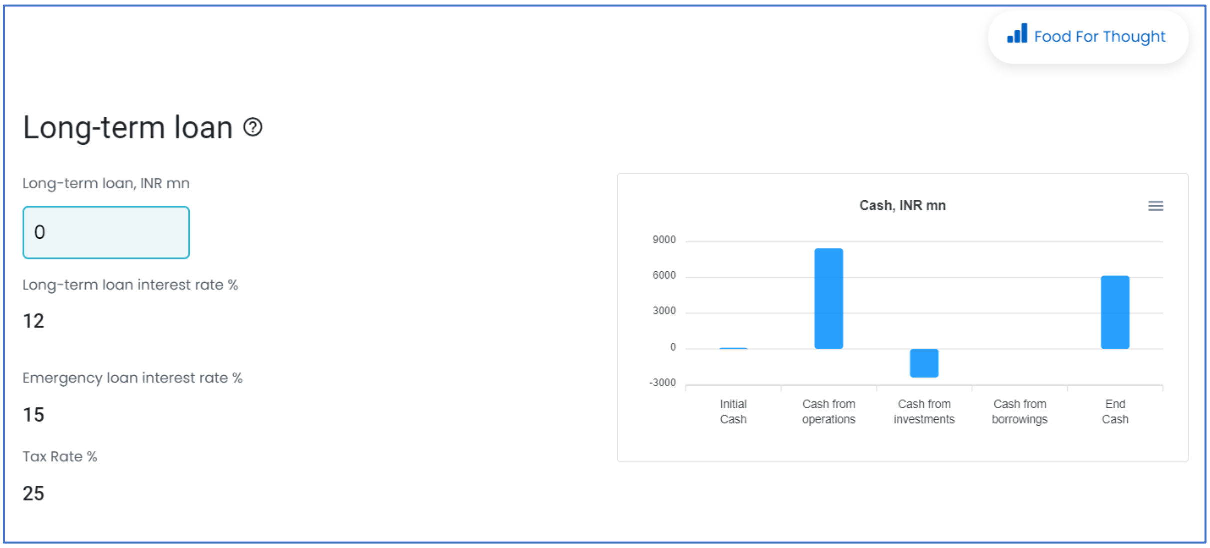

Figure 17 The Finance decisions in Game Arena

2.8 Decision checklist

The Decision Checklist helps participants review and finalize their decisions. Once decisions are confirmed, they can submit them using the "Submit" option and choose save and proceed. It's important to remember that once decisions are submitted and saved, they cannot be changed for that round. This ensures that decisions are carefully considered before finalization, promoting clarity and accountability in the decision-making process.

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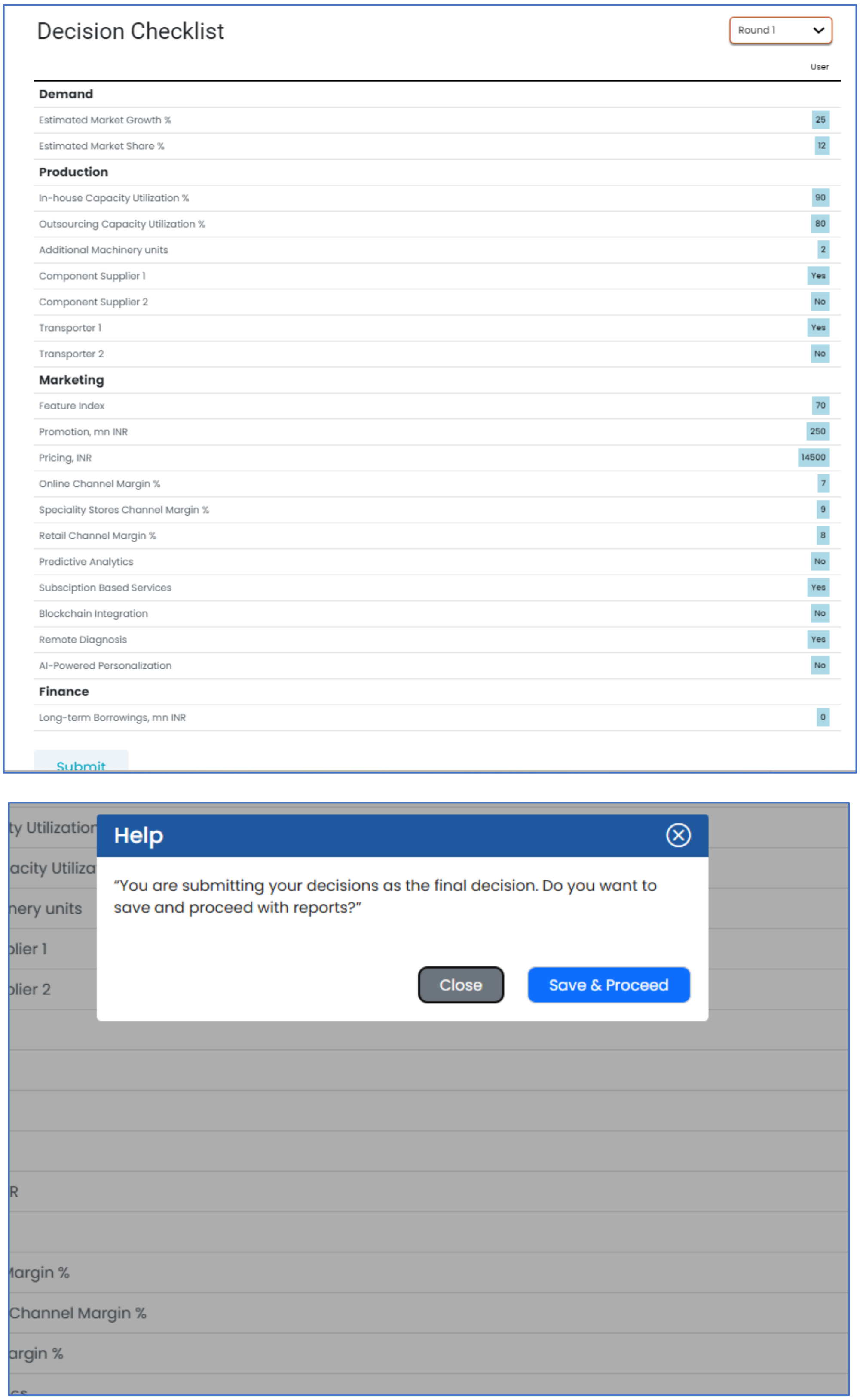

Figure 18 The Decision Checklist

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2.9 Report

The report section provides the details on the participants' critical thinking abilities in each aspect. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section consists of feedback on the performance of the participant based on four metrics.

- Rigour: Consistency throughout the decision-making and recommendation.
- Structuring: Logical approach of gathering and analysing information
- Synthesis: Approach towards interpreting information and making informed decisions
- Business Judgement: Ability to evaluate facts, and consider risks and possible implications when making decisions.

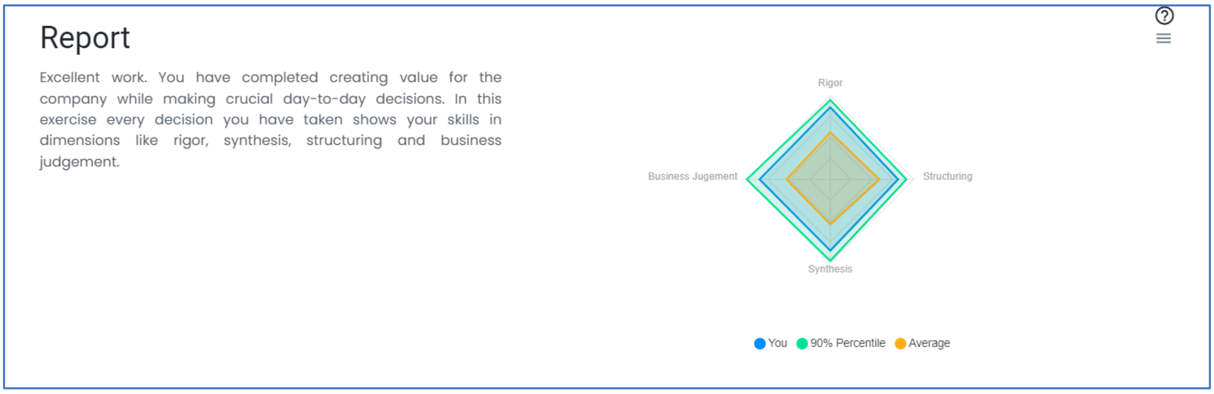


Figure 19 The Report Section

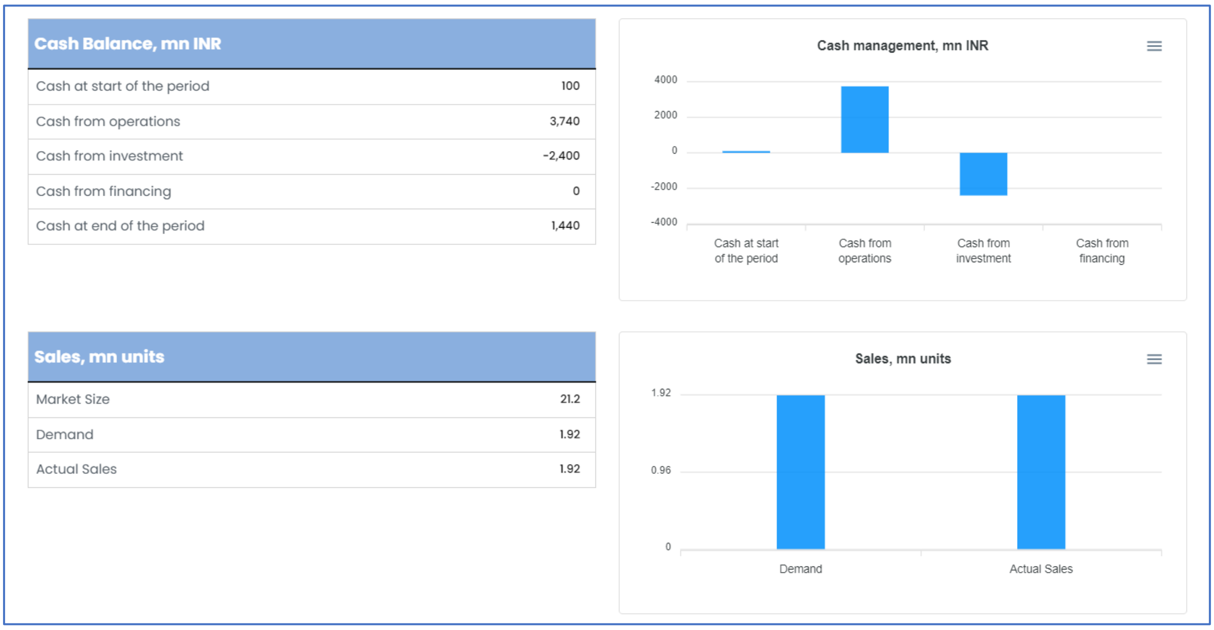
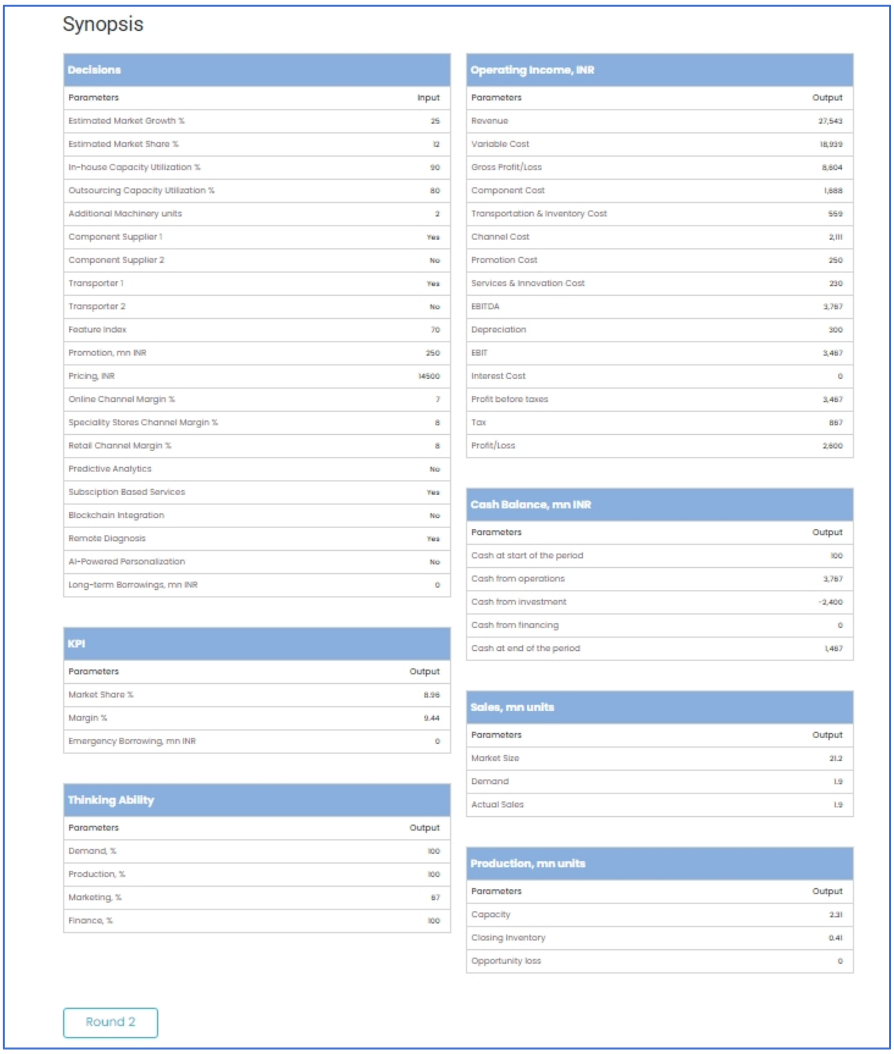


Figure 20 The Report Section

The report section offers an analysis of how decisions have influenced market share, profitability, and sales relative to demand. It also provides detailed insights into cash management. Key performance indicators (KPIs) for the simulation include achieving a market share of 8.3% or higher and maintaining an operating margin exceeding 10%. These metrics are critical benchmarks for evaluating the impact and success of decisions made throughout the simulation.

2.10 Synopsis

The synopsis section offers a concise summary of key decision areas and outcomes, highlighting critical insights including the KPI achieved and strategic directions taken during the simulation, facilitating comprehensive understanding and future planning.



Synopsis

Decisions	
Parameters	Input
Estimated Market Growth %	25
Estimated Market Share %	12
In-house Capacity Utilization %	90
Outsourcing Capacity Utilization %	80
Additional Machinery units	2
Component Supplier 1	Yes
Component Supplier 2	No
Transporter 1	Yes
Transporter 2	No
Feature Index	70
Promotion, mn INR	250
Pricing, INR	14500
Online Channel Margin %	7
Speciality Stores Channel Margin %	8
Retail Channel Margin %	8
Predictive Analytics	No
Subsciption Based Services	Yes
Blockchain Integration	No
Remote Diagnosis	Yes
AI-Powered Personalization	No
Long-term Borrowings, mn INR	0

KPI	
Parameters	Output
Market Share %	8.96
Margin %	9.44
Emergency Borrowing, mn INR	0

Thinking Ability	
Parameters	Output
Demand, %	100
Production, %	100
Marketing, %	67
Finance, %	100

Operating Income, INR	
Parameters	Output
Revenue	27,543
Variable Cost	18,939
Gross Profit/Loss	8,604
Component Cost	1,688
Transportation & Inventory Cost	559
Channel Cost	2,111
Promotion Cost	250
Services & Innovation Cost	230
EBITDA	3,787
Depreciation	300
EBIT	3,487
Interest Cost	0
Profit before taxes	3,487
Tax	887
Profit/Loss	2,600

Cash Balance, mn INR	
Parameters	Output
Cash at start of the period	100
Cash from operations	3,787
Cash from investment	-2,400
Cash from financing	0
Cash at end of the period	1,487

Sales, mn units	
Parameters	Output
Market Size	21.2
Demand	1.9
Actual Sales	1.9

Production, mn units	
Parameters	Output
Capacity	2.31
Closing Inventory	0.41
Opportunity loss	0

Round 2

Figure 21 The Synopsis Section

To proceed to the next round, click on the “next round” at the bottom of the synopsis section. This will redirect participants to the home page, where they can access the next round by clicking on the corresponding card.